LLMs Can Teach Themselves to Better Predict the Future

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Abstract

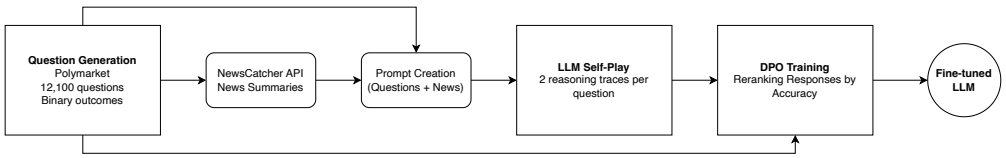
We present an outcome-driven fine-tuning framework that enhances the forecasting capabilities of large language models (LLMs) without relying on human-curated reasoning samples. Our method leverages model self-play to generate pairs of diverse reasoning trajectories and probabilistic forecasts for a set of diverse questions that resolve after the models' knowledge cutoff date. We then rank pairs of these reasoning traces by their distance to the actual outcomes before fine-tuning the model via Direct Preference Optimization (DPO). On a separate test set, our approach increases prediction accuracy of Phi-4 14B and DeepSeek-R1 14B by between 7–10% over a base model and a DPO fine-tuned control model with randomized labels, bringing them on par with forecasting capabilities of much larger frontier models like GPT-4o.

1 Introduction

Large language models (LLMs) have demonstrated remarkable capabilities in a wide range of areas, often approaching or exceeding human performance. One area where human performance has not yet been surpassed is judgemental forecasting [1], where probabilistic forecasts are assigned to future events. Successful forecasts by top-performing human forecasters include substantial reasoning about facts of the world, various trends, and competing pieces of evidence [2], making it a great place to study model reasoning capabilities in a messy real-world environment.

Moreover, forecasting is a central task in decision-making across sectors as diverse as finance, policy, and law. It is central to inform resource allocation, manage risks, and plan organizational decisions. Modern LLMs have already been shown to conduct financial analysis [3], evaluate the impact of events on time series [4], and improve climate policy decision-making [5]. This makes improving LLMs' forecasting abilities potentially impactful and wide-ranging.

Figure 1: Overview Flowchart



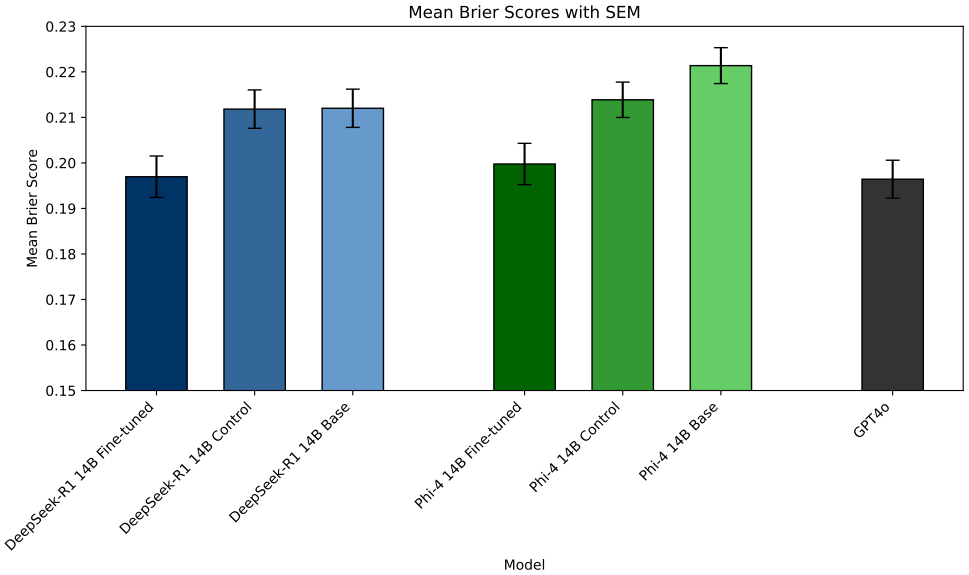
Notes: This chart outlines the stages of our method.

There has been some work explicitly looking to apply and boost the forecasting capabilities of LLMs. Such work has relied on aggregation [6], retrieval of news as well as fine-tuning [7], and ranked-based context retrieval [8], among other approaches [9]. While most of these systems improve performance to varying degrees, many share a common methodological limitation: They are frequently reliant on human-curated data such as up-to-date crowd forecasts or output curation, and often fail to have the models learn from resolved outcomes. Human outputs are slow and costly to procure, making it difficult to have models continually learn from them and improve.

In this paper, we propose a new approach to improving LLM forecasting performance that sidesteps the use of human inputs above and beyond real-world resolutions and enables the model to directly learn from

actual outcomes and self-play. Self-play, where models compete against themselves, has previously been used in AlphaGo Zero to achieve superhuman performance [10], as well as more recent fine-tuning approaches like Self-Play fIne-tuNing (SPIN) [11]. By allowing the model to produce reasonings and forecasts by itself on a large number of forecasting questions, this provides us with a large data set that we can then use for further training. As such, we do not rely on human-written forecasting rationales or predictions and instead only use model-generated reasoning, making this straightforwardly scalable.

Figure 2: Accuracy Results for all Models



Notes: The bar graph shows mean Brier scores with standard error of the mean (SEM) error bars. The y-axis starts at 0.15. Lower Brier scores indicate higher accuracy.

Further, our approach uses Direct Preference Optimisation (DPO) [12], a reward-free method entirely bypassing the need for a separate reward model, to instead learn a reward signal from sets of ranked reasoning pairs [13] drawn from the self-play outputs. This allows DPO to capitalize on relative rankings between forecasts, enabling the model to learn from the entire set of generated samples without the need for manual curation. Even when forecasts are individually suboptimal, DPO trains the model to discern subtle differences in quality and systematically correct biases through pairwise comparisons. By contrast, Supervised Fine-Tuning (SFT) relies on human-curated examples and treats selected forecasts as fully correct, which can lead to the discarding of potentially valuable information; DPO overcomes this limitation by learning from all samples, thereby enhancing the robustness and efficiency of the fine-tuning process.

Our work follows up on recent advances made by DeepSeek's release of R1 [14], which demonstrated the power of reinforcement learning in deterministic contexts like mathematics and coding. We move the focus to real-world forecasting, which is inherently noisy and relies on calibrated predictions rather than simple binary correctness. This requires our models to learn from noisy probabilistic outcomes, which, if successful, promises widespread applicability.

To achieve this, we draw on a large dataset of resolved prediction market questions from Polymarket, where the model—restricted to a historical cutoff date—generates multiple reasoning traces and probabilistic forecasts through self-play. We then rank these pairs of rationales based on their proximity to the resolved outcome (for instance, ranking a 5% prediction higher than a 10% prediction if the event resolved to "No") before fine-tuning our model on them and testing the model on a separate test set. This ensures that the model does not simply learn whether a forecast predicted that an event would or would not occur, but instead enables it to draw directly from the full set of forecasts needed for a well-calibrated forecasting model (see Figure 1).

Our results on a temporally held-out test set of questions resolving after December 25, 2024 show that

for both of the models that we employed our method on, Phi-4 14B [15] and DeepSeek-R1 14B [14], we find accuracy improvements of between 7–10% over the base versions of these models as well as the same models fine-tuned with randomized outcome labels as a control, see Figure 2. Comparing not only to the base model but also to randomized-label fine-tuned controls allows us to more rigorously tease out the effect of outcome-based learning versus exposure to additional information. This shows that our method improves model forecasting performance, underscoring the potential of drawing on self-play reasoning data in improving probabilistic reasoning and prediction accuracy. Strikingly, our fine-tune of both models are also on par with the performance of the much larger GPT-4o [16].

2 Method

Our approach consists of six main steps: 2.1) Collection and preprocessing of forecasting data, 2.2) News collection, 2.3) Synthetic training data generation through base model self-play, 2.4) Resolution-driven re-ranking, 2.5) Direct Preference Optimization (DPO) fine-tuning, and 2.6) Forecasting test-set questions.

For this pipeline, we used two models for self-play and for the final forecasting process: Phi-4 [15] and DeepSeek-R1-Distill-Qwen-14B [14]. Both models are small (at 14B parameters) but have shown strong performance on general science and coding benchmarks, sometimes rivalling GPT-4o [17, 18, 14]. DeepSeek-R1-Distill-Qwen-14B is a distilled model derived from Gwen2.5-14B [19] fine-tuned with the reasoning patterns from DeepSeek-R1 [14]. Throughout this paper, we refer to these models as Phi-4 14B and DeepSeek-R1 14B respectively.

2.1 Data

We collected a total of 12,100 binary outcome forecasting questions from the prediction market Polymarket. We excluded all outcomes with ambiguous resolutions and partitioned the data as follows: our training set included 9,800 questions that all resolved between July 1 and December 15, 2024, and our test set included 2,300 questions that all resolved between December 25, 2024 and January 23, 2025. We also collected the final outcomes for all questions, recording as '0' all outcomes that did not happen and as '1' all outcomes that did happen. See Table 1 for example questions.

Table 1: Example Questions with Outcomes

Question	**Outcome**
Will Sam Altman attend presidential inauguration?	1
FTX doesn't start payouts in 2024?	1
Will Modi win reelection?	1
Will Sebastian Korda reach the quarterfinals of the Australian Open?	0
Pershing Square IPO in 2024?	0

Notes: Table 1 shows a set of questions in the test set as well as their outcomes, with 0 indicating a negative resolution and 1 a positive resolution.

To evaluate the accuracy of our probabilistic forecasts in this paper, we calculate Brier scores. For each forecasting question with a predicted probability p_i and an actual outcome $o_i \in \{0, 1\}$, the Brier score is defined as

$$BS = \frac{1}{N} \sum (p_i - o_i)^2,$$

where N is the total number of forecasting questions. A lower Brier score indicates higher forecasting accuracy.